

Optimizing QAOA: An Empirical Study of Circuit Depth, Classical Optimizers, and Parameter Initialization Strategies for the Max-Cut Problem

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Abstract—The Quantum Approximate Optimization Algorithm (QAOA) is one of the most promising variational algorithms for combinatorial optimization on Noisy Intermediate-Scale Quantum (NISQ) devices, yet its performance depends critically on three practical choices: circuit depth p , the classical optimizer used to tune the variational parameters (γ, β) , and the parameter initialization strategy. This paper presents a systematic empirical study of these three factors for the Maximum Cut (Max-Cut) problem on three graph families (3-Regular, Erdős-Rényi, and Barabási-Albert) of sizes $n \in \{8, 10, 12\}$. We show that increasing p from 1 to 4 consistently improves the approximation ratio under gradient-free optimization (COBYLA), reaching up to 0.974 on 3-Regular graphs, while the gradient-based AdaGrad optimizer degrades at $p = 4$ on the largest instances, a signature of the barren-plateau phenomenon. A comparison of six classical optimizers at $p = 2$ shows that all deterministic methods (L-BFGS-B, COBYLA, AdaGrad) reach identical solution quality, with L-BFGS-B up to $20\times$ faster than AdaGrad. Among four initialization strategies, Fixed and Warm-Start initializations consistently outperform Random and Trotterized-Quantum-Annealing (TQA) initialization. Finally, a complete visualization of the $p = 1$ optimization landscape reveals 10 local minima organized in three quality tiers, and the number of local minima grows nearly five-fold to at least 49 at $p = 2$, which explains the empirical trends observed for depth, optimizer choice, and initialization. Based on these findings we derive practical guidelines for deploying QAOA on near-term devices.

Keywords—Quantum Approximate Optimization Algorithm (QAOA), variational quantum algorithms, combinatorial optimization, Maximum Cut (Max-Cut), parameter initialization, classical optimizers, barren plateaus, NISQ devices.

I. INTRODUCTION

Combinatorial optimization problems are ubiquitous in logistics, scheduling, network design, and finance, and many of them are NP-hard, so that classical exact and approximate methods reach their limits as instance size grows. Quantum computing offers an alternative computational paradigm for such problems. Among the algorithms proposed for near-term devices, the Quantum Approximate Optimization Algorithm (QAOA), introduced by Farhi, Goldstone, and Gutmann [1], encodes a combinatorial problem into a parameterized quantum circuit and relies on a classical optimizer to tune the circuit parameters in a hybrid quantum-classical loop. This makes

QAOA one of the most hardware-friendly quantum algorithms for current Noisy Intermediate-Scale Quantum (NISQ) devices [2].

Despite this promise, QAOA’s practical performance is highly sensitive to the circuit depth p , the choice of classical optimizer, and the parameter initialization strategy, and finding good parameters is expensive, often requiring hundreds of noisy circuit evaluations over a non-convex landscape riddled with local minima and, for larger systems, barren plateaus [3]. Understanding how these three factors interact is therefore essential for making QAOA usable in practice.

This paper reports a systematic empirical study addressing four objectives: (i) quantify the effect of circuit depth $p \in \{1, 2, 3, 4\}$ on solution quality and cost; (ii) compare six classical optimizers spanning gradient-based, gradient-free, and population-based families; (iii) evaluate four parameter initialization strategies; and (iv) characterize the QAOA optimization landscape to explain the observed empirical behavior. The study targets the Max-Cut problem on nine graph instances spanning three structural families and three sizes. The main contributions are: a controlled comparison of depth, optimizer, and initialization effects under identical experimental conditions; a full grid-based visualization of the $p = 1$ landscape and a cross-sectional analysis at $p = 2$; and a set of practical recommendations for QAOA deployment on small to medium NISQ-scale instances.

The remainder of the paper is organized as follows. Section II reviews related work. Section III presents the QAOA formulation. Section IV describes the experimental setup. Section V reports the results on depth, optimizers, and initialization. Section VI analyzes the optimization landscape. Section VII discusses the findings and gives practical guidelines, and Section VIII concludes with limitations and future work.

II. RELATED WORK

QAOA was introduced by Farhi, Goldstone, and Gutmann [1] together with a proof that for 3-Regular graphs, a single layer ($p = 1$) already guarantees an approximation ratio

of at least 0.6924 for Max-Cut. Its connection to quantum adiabatic computation [4] via Trotterization [5], [6] explains why deeper circuits should, in principle, approach the adiabatic optimum as $p \rightarrow \infty$. Hadfield *et al.* [7] generalized the algorithm to a Quantum Alternating Operator Ansatz with problem-specific mixers for constrained optimization. On the classical side, Goemans and Williamson’s semidefinite-programming relaxation [8] remains the strongest known classical approximation guarantee for Max-Cut (≈ 0.878) and is a useful reference point for QAOA’s empirical ratios.

A substantial body of work has studied why QAOA training is hard. McClean *et al.* [3] showed that randomly initialized variational circuits exhibit *barren plateaus*, where gradients vanish exponentially with qubit count, a result later connected to the QAOA landscape and to symmetry structure by Bravyi *et al.* [9]. Brandao *et al.* [10] proved that for fixed control parameters the QAOA objective concentrates across typical problem instances, motivating parameter-transfer strategies. Zhou *et al.* [11] analyzed QAOA performance and parameter patterns at larger p and proposed interpolation-based initialization schemes, which directly motivate the Warm-Start strategy evaluated in this paper. Shaydulin *et al.* [12] studied parameter transfer across weighted Max-Cut instances, showing that near-optimal parameters found on one instance often generalize to others of similar structure.

Relative to this literature, most existing studies isolate a single factor (depth, or optimizer, or initialization) or a single graph family. The present work instead evaluates all three factors under identical conditions across three structurally distinct graph families and links the empirical trends to a direct, grid-based characterization of the underlying optimization landscape, providing a mechanistic explanation rather than a purely empirical one.

III. QAOA BACKGROUND

QAOA can be derived as a Trotterized approximation of quantum adiabatic computation [4]. In the adiabatic algorithm, a system is prepared in the ground state of an initial Hamiltonian H_0 and evolved slowly toward a problem Hamiltonian H_P whose ground state encodes the optimal solution. Applying a first-order Suzuki–Trotter decomposition [5], [6] to this evolution and treating the resulting time-step coefficients as free variational parameters (γ_k, β_k) yields the QAOA unitary

$$U_{\text{QAOA}} = \prod_{k=1}^p e^{-i\beta_k H_0} e^{-i\gamma_k H_P}, \quad (1)$$

i.e., p alternating layers of a *cost* unitary generated by H_P and a *mixer* unitary generated by H_0 . For finite p , QAOA is a variational approximation of the adiabatic evolution, and p controls the trade-off between approximation quality and circuit cost.

For Max-Cut on a graph $G = (V, E)$ (Fig. 1), each binary variable $z_i \in \{0, 1\}$ is mapped to a qubit via $z_i \mapsto (1 - \sigma_i^z)/2$, and the cost Hamiltonian is

$$H_C = \sum_{(i,j) \in E} \frac{1}{2} (1 - \sigma_i^z \sigma_j^z), \quad (2)$$

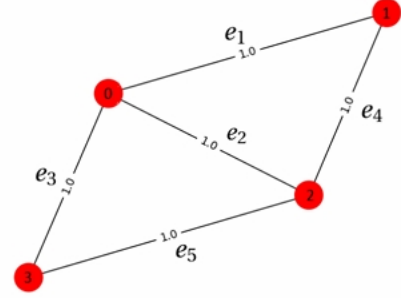


Fig. 1: Example Max-Cut problem instance: an unweighted graph $G = (V, E)$ to be partitioned into two sets so as to maximize the number of edges cut.

whose expectation value equals the cut size for a given bitstring. The standard transverse-field mixer $H_B = \sum_i \sigma_i^x$ drives transitions between basis states. Starting from the uniform superposition $|+\rangle^{\otimes n}$, the QAOA state is

$$|\gamma, \beta\rangle = \prod_{k=1}^p e^{-i\beta_k H_B} e^{-i\gamma_k H_C} |+\rangle^{\otimes n}, \quad (3)$$

and a classical optimizer iteratively updates (γ, β) to maximize $\langle H_C \rangle$. Performance is measured with the *approximation ratio*

$$r = \frac{C(z^*)}{C^*}, \quad (4)$$

where $C(z^*)$ is the best cut found and C^* the true optimum (obtained here by brute force since $n \leq 12$).

Equivalently, Max-Cut can be written as an Ising model over $\{-1, +1\}$ spins $s_i = 1 - 2z_i$, with cost

$$C(s) = \frac{1}{2} \sum_{(i,j) \in E} (1 - s_i s_j), \quad (5)$$

or as a Quadratic Unconstrained Binary Optimization (QUBO) problem over $z_i \in \{0, 1\}$; both formulations are equivalent up to an affine change of variables and both map directly onto the cost Hamiltonian H_C used by the quantum circuit. On 3-Regular graphs, QAOA at $p = 1$ is guaranteed to achieve an approximation ratio of at least 0.6924 [1], and this bound improves monotonically as p increases, approaching 1 as $p \rightarrow \infty$ in the adiabatic limit – the theoretical basis for the depth study in Section VA.

QAOA training is complicated by *barren plateaus*: regions where $\text{Var}[\partial F_p / \partial \theta_k] \leq O(1/2^n)$, so the gradient vanishes exponentially with system size [3], and by a landscape containing multiple local minima and basins of attraction whose number grows with both p and n . Algorithm 1 summarizes the resulting hybrid quantum–classical optimization loop used throughout this study.

IV. EXPERIMENTAL SETUP

A. Implementation

All experiments are implemented in Python using PennyLane 0.44.0 [13] with the `default.qubit` noiseless statevector simulator. Graph generation uses NetworkX 3.1 [14]

Algorithm 1 QAOA Optimization Loop for Max-Cut

Require: Graph $G = (V, E)$, depth p , initial $\theta_0 = (\gamma_0, \beta_0)$, max iterations T

Ensure: Optimized parameters θ^* , approximation ratio r^*

```
1: Build  $H_C = \sum_{(i,j) \in E} \frac{1}{2}(1 - \sigma_i^z \sigma_j^z)$ ;  $\theta \leftarrow \theta_0$ 
2: for  $t = 1$  to  $T$  do
3:   Prepare  $|\psi_0\rangle \leftarrow |+\rangle^{\otimes |V|}$  via Hadamard gates
4:   for  $k = 1$  to  $p$  do
5:     Apply  $U_C(\gamma_k) = e^{-i\gamma_k H_C}$ , then  $U_B(\beta_k) = e^{-i\beta_k H_B}$ 
6:   end for
7:   Measure  $\langle \psi(\theta) | H_C | \psi(\theta) \rangle$ ; evaluate objective  $f(\theta)$ 
8:   if convergence criterion met then break
9:   end if
10:   $\theta \leftarrow \text{ClassicalOptimizerStep}(f, \theta)$ 
11: end for
12:  $\theta^* \leftarrow \theta$ ;  $r^* \leftarrow -f(\theta^*)/C^*$ 
13: return  $\theta^*, r^*$ 
```

and numerical routines use NumPy 1.26.4. A fixed random seed (`np.random.seed(42)`) ensures reproducibility. Each circuit is executed with 100 measurement shots per evaluation, and the approximation ratio is computed against C^* obtained by exhaustive brute-force search over all 2^n bitstrings.

B. Graph Instances

We evaluate QAOA on nine graph instances spanning three structurally distinct families:

3-Regular Random Graphs: Each vertex has degree exactly 3. This family is the standard benchmark of the original QAOA paper [1], where the 0.6924 analytical bound was derived.

Erdős-Rényi Graphs: Each edge is included independently with probability $p = 0.5$, yielding denser and more irregular graphs with heterogeneous degree distributions.

Barabási-Albert Graphs: Generated by preferential attachment with $m = 2$, producing scale-free graphs with hub vertices characteristic of real-world networks.

Table I summarizes the graph properties.

TABLE I: Graph instances used in experiments.

Graph Type	n=8	n=10	n=12
3-Regular	12 edges	15 edges	18 edges
Erdős-Rényi	13 edges	17 edges	23 edges
Barabási-Albert	12 edges	16 edges	19 edges

C. Classical Optimizers

We evaluate six classical optimizers representing three categories:

Gradient-based: L-BFGS-B (quasi-Newton with approximate Hessian, exact gradients via the parameter shift rule [15])

and AdaGrad (adaptive per-parameter learning rate, $\eta = 0.1$, 200 iterations).

Gradient-free: COBYLA (trust-region simplex method, 300 iterations, `rhobeg=0.5`) and SPSA (stochastic two-point gradient approximation, 200 iterations, $a = 0.16$, $c = 0.1$, $\alpha = 0.602$, $\gamma = 0.101$, 5 independent trials).

Population-based: Differential Evolution (DE, strategy `best1bin`, `popsiz=10`, 20 generations, mutation(0.5,1.0), `CR=0.7`, 5 trials) and a Hybrid DE+L-BFGS-B scheme (coarse DE search for 8 generations followed by L-BFGS-B local refinement for up to 100 iterations, 5 trials).

Stochastic methods (SPSA, DE, Hybrid) are reported as mean $\pm$ standard deviation over 5 independent trials. Deterministic methods (L-BFGS-B, COBYLA, AdaGrad) are run once. All optimizer comparisons are conducted at $p = 2$ ($2p = 4$ parameters).

D. Initialization Strategies

Four strategies are compared using L-BFGS-B as optimizer at $p \in \{1, 2\}$:

Random: $\gamma_k \sim \mathcal{U}(0, 2\pi)$, $\beta_k \sim \mathcal{U}(0, \pi)$; 10 independent trials.

Fixed: $\theta_k = 1.0$ for all parameters; 1 trial (deterministic).

TQA: γ_k increasing linearly from 0.1 to 1.0, β_k decreasing from 0.5 to 0.1, inspired by the adiabatic schedule; 1 trial.

Warm-Start: Parameters are initialized by interpolating from the best optimal parameters found at depth $p-1$, with a small Gaussian perturbation ($\sigma = 0.05$); 10 trials. Algorithm 2 details the procedure.

Algorithm 2 Warm-Start Parameter Initialization

Require: Depth p , best parameters θ_{p-1}^* from previous depth

Ensure: Initial parameters $\theta^{(0)} \in \mathbb{R}^{2p}$

```
1: if  $p = 1$  or  $\theta_{p-1}^*$  is None then
2:    $\theta^{(0)} \leftarrow \mathcal{U}(0, 2\pi)^{2p}$  ▷ Random fallback
3: else
4:    $p_{\text{prev}} \leftarrow |\theta_{p-1}^*|/2$ 
5:    $\gamma_{\text{prev}}^* \leftarrow \theta_{p-1}^*[0 : p_{\text{prev}}]$ 
6:    $\beta_{\text{prev}}^* \leftarrow \theta_{p-1}^*[p_{\text{prev}} : 2p_{\text{prev}}]$ 
7:   for  $k = 0, 1, \dots, p-1$  do
8:      $i \leftarrow \min(k, p_{\text{prev}} - 1)$ 
9:      $\gamma_k^{(0)} \leftarrow \gamma_{\text{prev}}^*[i] + \mathcal{N}(0, 0.05)$ 
10:     $\beta_k^{(0)} \leftarrow \beta_{\text{prev}}^*[i] + \mathcal{N}(0, 0.05)$ 
11:   end for
12: end if
13: return  $\theta^{(0)} = [\gamma_0^{(0)}, \dots, \gamma_{p-1}^{(0)}, \beta_0^{(0)}, \dots, \beta_{p-1}^{(0)}]$ 
```

V. RESULTS

A. Impact of Circuit Depth

Table II reports approximation ratios vs. depth p for COBYLA across all graph families and sizes. Figure 2 shows the COBYLA vs. AdaGrad comparison for 3-Regular graphs.

Three consistent patterns emerge. First, COBYLA achieves monotonically non-decreasing approximation ratios with p on

TABLE II: Approximation ratio vs. depth p (COBYLA).

Family	n	p=1	p=2	p=3	p=4
3-Regular	8	0.831	0.920	0.945	0.974
	10	0.799	0.875	0.905	0.945
	12	0.779	0.859	0.886	0.932
Erdős-Rényi	8	0.586	0.595	0.712	0.697
	10	0.607	0.609	0.772	0.777
	12	0.617	0.629	0.738	0.725
Barabási-Albert	8	0.706	0.733	0.814	0.819
	10	0.691	0.728	0.801	0.811
	12	0.740	0.773	0.835	0.836

3-Regular and Barabási-Albert graphs, confirming the theoretical guarantee. Erdős-Rényi graphs show a decrease at $p = 4$ for $n = 8$ ($0.712 \rightarrow 0.697$) and $n = 12$ ($0.738 \rightarrow 0.725$), indicating that even gradient-free methods encounter increased landscape complexity on denser instances at higher depths. Second, graph structure matters: 3-Regular graphs consistently achieve the highest ratios owing to their uniform, symmetric structure, while Erdős-Rényi graphs are the hardest due to their irregular, denser topology. Barabási-Albert graphs fall in between. Third, AdaGrad (unlike COBYLA) shows degradation at $n = 12$, $p = 4$ on 3-Regular graphs ($0.9195 \rightarrow 0.9002$), attributable to an early, size-dependent onset of the barren plateau effect at this graph size and depth, as confirmed by the landscape analysis in Section VI.

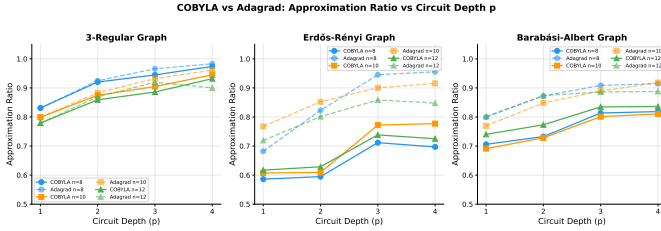


Fig. 2: Approximation ratio vs. circuit depth p for COBYLA (solid) and AdaGrad (dashed) on 3-Regular graphs. AdaGrad degrades at $n = 12$, $p = 4$ due to incipient barren plateaus, while COBYLA maintains monotonic improvement.

B. Comparison of Classical Optimizers

Table III compares all six optimizers at $p = 2$, $n = 12$ across all graph families. Figure 3 shows representative convergence curves on 3-Regular graphs.

At $p = 2$, the three deterministic optimizers reach *identical* approximation ratios on 3-Regular and, notably, L-BFGS-B and AdaGrad also match on the harder Erdős-Rényi and Barabási-Albert graphs, while COBYLA lags behind on these denser instances, indicating that gradient information becomes increasingly valuable as graph irregularity increases at this shallow depth. L-BFGS-B is the fastest deterministic method, up to $3\times$ faster than COBYLA and up to $20\times$ faster than AdaGrad, owing to its use of curvature information via an

TABLE III: Optimizer comparison at $p = 2$, $n = 12$ (approximation ratio). $\dagger$ Average time on 3-Regular. Stochastic methods report mean $\pm$ std over 5 trials.

Optimizer	3-Reg.	Erdős-R.	Barab.-Alb.	Time (s) †
L-BFGS-B	0.8697	0.8014	0.8735	4.49
COBYLA	0.8697	0.6288	0.8735	11.46
AdaGrad	0.8697	0.8014	0.8735	58.71
SPSA	$0.856 \pm .017$	—	—	52.86
DE	$0.861 \pm .015$	—	—	35.31
Hybrid	$0.855 \pm .009$	—	—	16.67

approximate Hessian. Stochastic methods (SPSA, DE, Hybrid) reach comparable mean quality but with non-negligible variance, most pronounced on denser graphs. The **Hybrid DE+L-BFGS-B scheme shows zero improvement** from its L-BFGS-B refinement stage in every configuration tested, indicating that the 8-generation DE coarse search is too short to leave the basin found by DE alone. Overall, L-BFGS-B is the most efficient choice at $p = 2$, with COBYLA as a gradient-free fallback for noisy hardware.

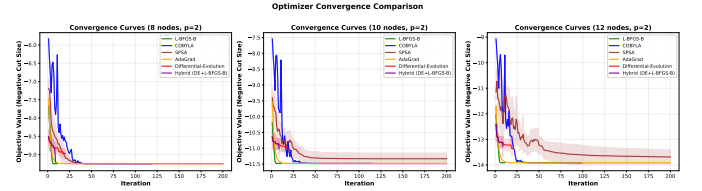


Fig. 3: Convergence curves of all six optimizers on 3-Regular graphs at $p = 2$ (shaded bands: ± 1 std. dev. over 5 trials for stochastic methods). Deterministic methods converge smoothly to the shared optimum; stochastic methods show wider spread.

C. Parameter Initialization Strategies

Table IV reports initialization strategy results for 3-Regular graphs using L-BFGS-B at $p \in \{1, 2\}$.

At $p = 1$, all strategies converge to essentially the same solution, since the two-parameter landscape has a single dominant basin. At $p = 2$, Fixed initialization achieves the best or tied-best approximation ratio across all graph types and sizes — a somewhat surprising result explained in Section VI — while Warm-Start achieves statistically indistinguishable quality in noticeably fewer iterations by reusing the optimum found at $p - 1$. TQA consistently underperforms Fixed and Warm-Start, indicating the linear adiabatic schedule is not a reliable practical guide, and Random initialization shows the largest variance, especially on Erdős-Rényi graphs (std up to 0.102 at $n = 10$).

VI. OPTIMIZATION LANDSCAPE ANALYSIS

To explain the results above, the $p = 1$ landscape of the 8-node 3-Regular instance is fully mapped on a 100×100 grid over $\gamma \in [0, 2\pi)$, $\beta \in [0, \pi)$, and the $p =$

TABLE IV: Initialization strategies, 3-Regular graphs, L-BFGS-B optimizer. Stochastic strategies (Random, Warm-Start) report mean $\pm$ std over 10 trials.

n	Strategy	Ratio (mean $\pm$ std)	Time (s)	Iters
$p = 1$				
8	Random	0.831 $\pm$ 0.000	0.62	31
8	Fixed	0.831 $\pm$ 0.000	0.65	30
8	TQA	0.831 $\pm$ 0.000	0.64	30
8	Warm-Start	0.808 $\pm$ 0.069	0.66	34
$p = 2$				
8	Random	0.907 $\pm$ 0.014	4.50	108
8	Fixed	0.925 $\pm$ 0.000	2.42	80
8	TQA	0.893 $\pm$ 0.000	3.16	110
8	Warm-Start	0.925 $\pm$ 0.000	1.78	59
12	Random	0.855 $\pm$ 0.012	5.94	105
12	Fixed	0.870 $\pm$ 0.000	4.19	75
12	TQA	0.837 $\pm$ 0.000	5.08	90
12	Warm-Start	0.864 $\pm$ 0.010	5.07	88

2 landscape (4 parameters) is examined through 2D and 1D cross-sections around the optimum $(\gamma_1^*, \gamma_2^*, \beta_1^*, \beta_2^*) = (0.449, 0.934, 1.130, 1.333)$, which achieves $r = 0.837$ versus 0.778 at $p = 1$. Table V summarizes landscape complexity metrics across depths.

TABLE V: QAOA landscape complexity vs. circuit depth ($n = 8$, 3-Regular).

p	Params	Local Minima	Mean $ \nabla f $	Barren (%)
1	2	10	2.439	0.8
2	4	≥ 49	3.406	0.0
3	6	≥ 50	4.242	0.0
4	8	≥ 50	3.829	0.0

The $p = 1$ landscape contains exactly 10 local minima organized in three quality tiers (Fig. 4), summarized as a growth trend in Fig. 5: the Fixed initialization point (1.0, 1.0) starts at a cut value of 6.51/10 and lies inside a basin leading to a top-tier minimum, which explains why Fixed initialization performs so well in Table IV, while random starting points reach the best-quality tier in only about 40% of trials, explaining Random’s variance. Between $p = 1$ and $p = 2$, the number of local minima grows nearly five-fold (from 10 to ≥ 49), directly explaining the increased optimization variance observed for deeper circuits in Section V-A. The 1D and 2D cross-sections of the $p = 2$ landscape show clear but non-convex optima along each parameter axis. For $n = 8$, mean gradient magnitude remains strong across all tested depths (barren-plateau coverage $\leq 0.8\%$), consistent with the theoretical expectation that barren plateaus become dominant only for much larger systems ($n \gtrsim 20$) [3]; the AdaGrad degradation observed at $n = 12$, $p = 4$ in Section V-A is therefore attributable to an early, size-dependent onset of this effect rather than a landscape property already present at $n = 8$.

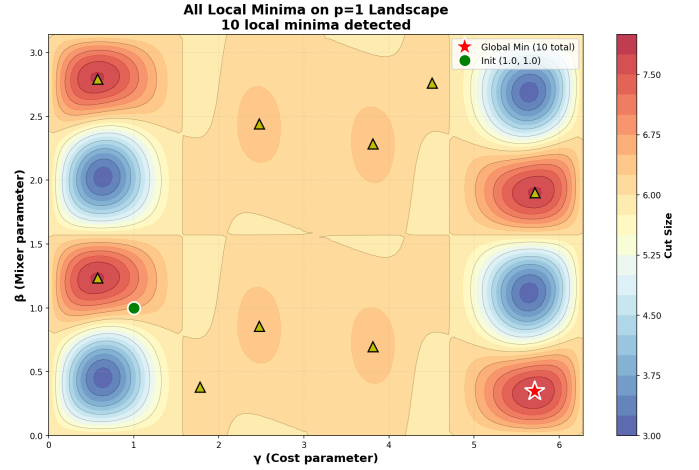


Fig. 4: All local minima detected on the $p = 1$ landscape (8-node 3-Regular instance), colored by quality tier (Δ = local minimum, $\star$ = global optimum, $\bullet$ = Fixed initialization point (1.0, 1.0)). The Fixed point falls within the basin of a top-tier minimum.

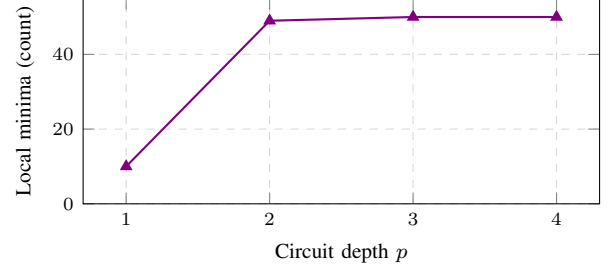


Fig. 5: Growth of the number of detected local minima with circuit depth p (8-node 3-Regular instance). The near five-fold jump between $p = 1$ and $p = 2$ explains the increased optimization variance observed for deeper circuits.

VII. DISCUSSION AND PRACTICAL RECOMMENDATIONS

Three consistent patterns emerge. First, *circuit depth is the primary lever for solution quality*, but its benefit is optimizer-dependent: gradient-free COBYLA scales reliably with p while gradient-based AdaGrad is vulnerable to the growing number of local minima and incipient barren plateaus at larger n and p . Second, *graph structure matters as much as algorithmic choices*: uniform 3-Regular graphs are consistently easiest for QAOA, Barabási–Albert graphs are intermediate, and denser, irregular Erdős–Rényi graphs are hardest, so QAOA performance should not be assumed to transfer across graph families. Third, *initialization and optimizer choice interact with the landscape’s basin structure*: Fixed and Warm-Start initialization succeed because they systematically land inside high-quality basins, and L-BFGS-B is efficient because strong gradients still dominate the tested regime.

Based on the combined evidence, we recommend: (1) using $p = 3$ or $p = 4$ for small-to-medium instances ($n \leq 12$) to balance quality and cost; (2) preferring COBYLA for

robustness across graph types and depths, especially where gradient reliability is a concern (e.g., on noisy hardware), with L-BFGS-B as the efficient choice when gradients are available and the instance is small; (3) using Warm-Start when optimizing across multiple depths, and Fixed initialization ($\theta_k = 1.0$) for single-depth optimization; and (4) avoiding TQA initialization on irregular graphs, where the adiabatic schedule is a poor proxy for the true optimum.

These recommendations should be read against the scope of the study. All experiments use noiseless statevector simulation, so the reported approximation ratios represent an upper bound on what current NISQ hardware, subject to decoherence, gate errors, and measurement noise, would achieve in practice. Similarly, the barren-plateau effect is observed only in its early, size-dependent onset at $n = 12$; its severity is expected to grow substantially for the larger qubit counts relevant to industrially meaningful Max-Cut instances, where the gap between gradient-based and gradient-free optimizers documented here is likely to widen further. Finally, because the ranking of optimizers and initialization strategies is shown to depend on graph structure, practitioners working with graph families not covered here (e.g., highly clustered or weighted graphs) should treat the specific numerical thresholds as indicative rather than universal, while the qualitative mechanism – basin structure explaining initialization performance, and local-minima growth explaining depth- and optimizer-dependent behavior – is expected to generalize.

VIII. CONCLUSION

This paper presented a systematic empirical and landscape-theoretic study of QAOA parameter optimization for Max-Cut. Increasing circuit depth from $p = 1$ to $p = 4$ consistently improves solution quality under gradient-free optimization (up to $r = 0.974$ on 3-Regular graphs), while gradient-based AdaGrad degrades at $p = 4$ on larger instances due to the barren-plateau phenomenon. At $p = 2$, all deterministic optimizers reach equivalent quality, with L-BFGS-B up to $20\times$ faster than AdaGrad, and Fixed and Warm-Start initialization consistently outperform Random and TQA. A full landscape characterization shows that the number of local minima grows nearly five-fold from $p = 1$ (10 minima) to $p = 2$ (≥ 49 minima), providing a rigorous explanation for all three empirical trends.

The main limitations of this study are the small problem sizes ($n \leq 12$), the use of noiseless simulation, the focus on a single benchmark problem (Max-Cut), and the restriction to circuit depths up to $p = 4$. These limitations motivate several directions for future work: extending the study to larger instances ($n > 20$) where barren plateaus are expected to dominate; repeating the experiments on real NISQ hardware (e.g., IBM Quantum, IonQ) to quantify the effect of gate and readout noise; exploring learned or problem-graph-aware initialization strategies; applying the same experimental framework to other combinatorial problems such as Max-SAT, the Traveling Salesman Problem, or vehicle routing; and evaluating Recursive QAOA as a way to mitigate the local-minima

proliferation documented in Section VI. As quantum hardware scales, we expect this combination of controlled empirical comparison and landscape-level explanation to remain a useful methodology for guiding practical QAOA deployment.

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